MAT2040
Tutorial 09

TA Group
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July 3&4, 2024

Midterm Q2.

(a)

$$A \xrightarrow[P]{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 4 & 3 \\ 2 & 2 & 3 \end{bmatrix} \xrightarrow[E_1]{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 2 & 2 & 3 \end{bmatrix}$$

$$\xrightarrow[E_2]{R_3 \to R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix} = U$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Midterm Q2.

(a) (cont'd)

Given $E_2E_1PA = U$, we have $PA = E_1^{-1}E_2^{-1}U = LU$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

Midterm Q2.

(b) L and U are not unique, U can be multiplied by any Type II elementary matrices with conrresponding L.
For instance,

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{bmatrix} \xrightarrow[E_3]{R_3 \to -R_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = U$$

Given $E_3E_2E_1PA = U$, we have $PA = E_1^{-1}E_2^{-1}E_3^{-1}U = LU$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Midterm Q4.

Assume $\mathbf{x}$ is an arbitrary vector in $Null(A)$,

$$A\mathbf{x} = \mathbf{0} \Rightarrow A^T A\mathbf{x} = \mathbf{0}$$

Hence $\mathbf{x}$ is also in $Null(A^T A)$, and $Null(A) \subseteq Null(A^T A)$.

Assume $\mathbf{x}$ is an arbitrary vector in $Null(A^T A)$,

$$A^T A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x}^T A^T A\mathbf{x} = \mathbf{0} \Rightarrow (A\mathbf{x})^T A\mathbf{x} = \mathbf{0} \Rightarrow A\mathbf{x} = \mathbf{0}$$

Hence $\mathbf{x}$ is also in $Null(A)$, and $Null(A^T A) \subseteq Null(A)$.

$$\Rightarrow Null(A) = Null(A^T A) \Rightarrow \dim(Null(A)) = \dim(Null(A^T A))$$

By Rank-Nullity Theorem, $\text{rank}(A) = \text{rank}(A^T A)$.

Midterm Q5.

For $AB = C$, we can prove that $Col(C)$ is a subspace of $Col(A)$. Let $\mathbf{x}$ be an arbitrary column vector in $Col(C)$. Since $\mathbf{x}$ is a linear combination of columns of C, we have

$$\mathbf{x} = C\mathbf{v} = AB\mathbf{v} = A(B\mathbf{v}) = A\mathbf{u}$$

where $\mathbf{v}$ and $\mathbf{u}$ are some column vectors. Therefore, $\mathbf{x}$ also belongs to $Col(A)$.

Similarly, we can prove that $Row(C)$ is a subspace of $Row(B)$. Hence $\text{rank}(C) \leq \min(\text{rank}(A), \text{rank}(B))$

$$\text{rank}(I_m) = m \leq \min(m, n), \quad \text{rank}(I_n) = n \leq \min(m, n)$$

$$\Rightarrow m = n$$

Midterm Q6.

Let $\mathcal{B} = \left[\begin{array}{cccc} \mathbf{b_1}, & \mathbf{b_2}, & \ldots, & \mathbf{b_n} \end{array} \right]$, we can express $\mathbf{u}$ and $\mathbf{v}$ as:

$$[\mathbf{u}]_{\mathcal{B}} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \rightarrow \mathbf{u} = u_1\mathbf{b_1} + u_2\mathbf{b_2} + \cdots + u_n\mathbf{b_n}$$

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \rightarrow \mathbf{v} = v_1\mathbf{b_1} + v_2\mathbf{b_2} + \cdots + v_n\mathbf{b_n}$$

Midterm Q6.

Accoding to A5 and A6,

$$\begin{aligned}\alpha\mathbf{u} &= \alpha(u_1\mathbf{b_1} + u_2\mathbf{b_2} + \cdots + u_n\mathbf{b_n}) \\ &= \alpha(u_1\mathbf{b_1}) + \alpha(u_2\mathbf{b_2}) + \cdots + \alpha(u_n\mathbf{b_n}) \\ &= \alpha u_1\mathbf{b_1} + \alpha u_2\mathbf{b_2} + \cdots + \alpha u_n\mathbf{b_n}\end{aligned}$$

Similarly,

$$\beta\mathbf{v} = \beta v_1\mathbf{b_1} + \beta v_2\mathbf{b_2} + \cdots + \beta v_n\mathbf{b_n}$$

According to A2 and A7,

$$\begin{aligned}\alpha\mathbf{u} + \beta\mathbf{v} &= \alpha u_1\mathbf{b_1} + \cdots + \alpha u_n\mathbf{b_n} + \beta v_1\mathbf{b_1} + \cdots + \beta v_n\mathbf{b_n} \\ &= (\alpha u_1\mathbf{b_1} + \beta v_1\mathbf{b_1}) + \cdots + (\alpha u_n\mathbf{b_n} + \beta v_n\mathbf{b_n}) \\ &= (\alpha u_1 + \beta v_1)\mathbf{b_1} + \cdots + (\alpha u_n + \beta v_n)\mathbf{b_n}\end{aligned}$$

Midterm Q6.

Therefore,

$$[\alpha\mathbf{u}+\beta\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} \alpha u_1 + \beta v_1 \\ \alpha u_2 + \beta v_2 \\ \vdots \\ \alpha u_n + \beta v_n \end{bmatrix} = \alpha \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \beta \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \alpha[\mathbf{u}]_{\mathcal{B}} + \beta[\mathbf{v}]_{\mathcal{B}}$$